

1. Write a program to check whether a number is even or odd using switch-case.

Code:

```
package day_01;

import java.util.Scanner;

public class Even_odd {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter number: ");

        int num = sc.nextInt();

        switch (num % 2) {

            case 0: System.out.println("Even"); break;

            case 1: System.out.println("Odd"); break;

        }

    }

}
```

2. Write a program that takes a character as input and checks whether it is a vowel or consonant using switch-case.

Code:

```
package day_02;

import java.util.Scanner;

public class Vowel_Consonent {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter character: ");

        char ch = sc.next().charAt(0);

        switch (ch) {

            case 'a': case 'e': case 'i': case 'o': case 'u':

            case 'A': case 'E': case 'I': case 'O': case 'U':

                System.out.println("Vowel");

                break;

        }

    }

}
```

```

        default:
            System.out.println("Consonant");
    }
}
}
}

```

3. Write a program that takes a grade (A, B, C, D, F) and displays the performance message.

Code:

```

package day_02;

import java.util.Scanner;

public class Grades {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter grade: ");

        char grade = sc.next().charAt(0);

        switch (grade) {

            case 'A': System.out.println("Excellent"); break;

            case 'B': System.out.println("Good"); break;

            case 'C': System.out.println("Average"); break;

            case 'D': System.out.println("Below Average"); break;

            case 'F': System.out.println("Fail"); break;

            default: System.out.println("Invalid Grade");

        }

    }

}

```

4. Create a menu-driven program for ATM operations: • 1 → Check Balance • 2 → Deposit • 3 → Withdraw • 4 → Exit Perform the operation based on user choice.

Code:

```

package day_02;

```

```
import java.util.Scanner;
```

```
public class ATM {
```

```
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        int balance = 10000;
```

```
        System.out.println("1. Balance. ");
```

```
        System.out.println("2. Updated Balance after deposit. ");
```

```
        System.out.println("3. Remaining Balance after withdraw. ");
```

```
        System.out.println("4. Exit. ");
```

```
        System.out.print("Enter choice: ");
```

```
        int choice = sc.nextInt();
```

```
        switch (choice) {
```

```
            case 1:
```

```
                System.out.println("Balance: " + balance);
```

```
                break;
```

```
            case 2:
```

```
                System.out.println("Enter deposit ammount: ");
```

```
                int deposit = sc.nextInt();
```

```
                balance += deposit;
```

```
                System.out.println("Updated Balance: " + balance);
```

```
                break;
```

```
            case 3:
```

```
                System.out.println("Enter withdraw ammount: ");
```

```
                int withdraw = sc.nextInt();
```

```
                if (withdraw <= balance) {
```

```
                    balance -= withdraw;
```

```
                    System.out.println("Remaining Balance: " + balance);
```

```

        } else {
            System.out.println("Insufficient Balance");
        }
        break;
    case 4:
        System.out.println("Exit");
        break;
    default:
        System.out.println("Invalid Choice");
    }
}
}
}

```

5. Create a program where user selects food item number and the program displays item name and price.

Code:

```

package day_02;

import java.util.Scanner;

public class Menu_Driven {

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter item number (1-4): ");

        int item = sc.nextInt();

        switch (item) {
            case 1: System.out.println("Burger - 100"); break;
            case 2: System.out.println("Pizza - 200"); break;
            case 3: System.out.println("Pasta - 150"); break;
            case 4: System.out.println("Sandwich - 80"); break;
            default: System.out.println("Invalid Item");
        }
    }
}

```

```
    }  
}  
}
```

6. Write a program that converts currency based on user choice (e.g., INR to USD, INR to EUR, etc.).

Code:

```
package day_02;
```

```
import java.util.Scanner;
```

```
public class Currency_Converter {
```

```
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        System.out.println("1. USD. ");
```

```
        System.out.println("2. EUR. ");
```

```
        System.out.println("3. GBP. ");
```

```
        System.out.print("Enter choice: ");
```

```
        int choice = sc.nextInt();
```

```
        System.out.print("Enter amount: ");
```

```
        double amount = sc.nextDouble();
```

```
        switch (choice) {
```

```
            case 1: System.out.println("USD: " + (amount / 83)); break;
```

```
            case 2: System.out.println("EUR: " + (amount / 90)); break;
```

```
            case 3: System.out.println("GBP: " + (amount / 100)); break;
```

```
            default: System.out.println("Invalid Choice");
```

```
        }
```

```
    }
```

```
}
```

7. Create a program that checks user role (Admin, User, Guest) and displays access permissions.

Code:

```
package day_02;

import java.util.Scanner;

public class User_Role_Access {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.println("1. Admin ");

        System.out.println("2. User ");

        System.out.println("3. Guest ");

        System.out.print("Who are you? ");

        String role = sc.next();

        switch (role) {

            case "Admin": System.out.println("Full Access"); break;

            case "User": System.out.println("Limited Access"); break;

            case "Guest": System.out.println("View Only"); break;

            default: System.out.println("Invalid Role");

        }

    }

}
```

8. Write a program to calculate the factorial of a given number using a while loop.

Code:

```
package day_02;

import java.util.Scanner;

public class Factorial {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.println("Enter a number: ");

        int n = sc.nextInt();

        int fact = 1;
```

```

        while (n > 0) {
            fact *= n;
            n--;
        }
        System.out.println("Factorial: " + fact);
    }
}

```

9. Write a program to print the multiplication table of a given number using a while loop.

Code:

```

package day_02;

import java.util.Scanner;

public class Multiplication_table {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter a number: ");
        int num = sc.nextInt();
        int i = 1;
        while (i <= 10) {
            System.out.println(num + " x " + i + " = " + (num * i));
            i++;
        }
    }
}

```

10. Write a program to count the number of digits in a given number using a while loop.

Code:

```

package day_02;

import java.util.Scanner;

public class Count_Digits {
    public static void main(String[] args) {

```

```

Scanner sc = new Scanner(System.in);

System.out.print("Enter a number: ");

int num = sc.nextInt();

int count = 0;

while (num != 0) {
    num /= 10;
    count++;
}

System.out.println("Digits: " + count);
}
}

```

11. Write a program to reverse a given number using a while loop.

Code:

```

package day_02;

import java.util.Scanner;

public class Reverse_number {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a number: ");

        int num = sc.nextInt();

        int rev = 0;

        while (num != 0) {
            rev = rev * 10 + num % 10;
            num /= 10;
        }

        System.out.println("Reverse: " + rev);

    }

}

```

12. Write a program to check whether a number is a palindrome using a while loop.

Code:

```
package day_02;

import java.util.Scanner;

public class Palindrome {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a number: ");

        int num = sc.nextInt();

        int temp = num, rev = 0;

        while (num != 0) {

            rev = rev * 10 + num % 10;

            num /= 10;

        }

        if (temp == rev)

            System.out.println("Palindrome");

        else

            System.out.println("Not Palindrome");

    }

}
```

13. Write a program to find the sum of digits of a number using a while loop.

Code:

```
package day_02;

import java.util.Scanner;

public class SumOfDigits {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a number: ");

        int num = sc.nextInt();

        int sum = 0;

        while (num != 0) {
```

```

        sum += num % 10;

        num /= 10;
    }

    System.out.println("Sum: " + sum);
}
}

```

14. Write a program to check whether a number is an Armstrong number using a while loop.

Code:

```

package day_02;

import java.util.Scanner;

public class Armstrong {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a number: ");

        int num = sc.nextInt();

        int temp = num, sum = 0;

        while (num != 0) {

            int digit = num % 10;

            sum += digit * digit * digit;

            num /= 10;

        }

        if (sum == temp)

            System.out.println("Armstrong");

        else

            System.out.println("Not Armstrong");

    }

}

```

15. Write a program to check whether a number is prime using a while loop.

Code:

```

package day_02;

import java.util.Scanner;

public class PrimeNumber {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a number: ");

        int num = sc.nextInt();

        int i = 2;

        boolean isPrime = true;

        while (i <= num / 2) {

            if (num % i == 0) {

                isPrime = false;

                break;

            }

            i++;

        }

        if (isPrime && num > 1)

            System.out.println("Prime");

        else

            System.out.println("Not Prime");

    }

}

```

16. Write a program to print Fibonacci series up to N terms using a while loop.

Code:

```

package day_02;

import java.util.Scanner;

public class Fibonacci {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a number: ");

```

```
int n = sc.nextInt();

int a = 0, b = 1, i = 1;

while (i <= n) {

    System.out.print(a + " ");

    int next = a + b;

    a = b;

    b = next;

    i++;

}

}
```